

1 Introduction

Following the Supreme Court’s ruling in *Murphy v. National Collegiate Athletic Association* (NCAA) (2018), the Professional and Amateur Sports Protection Act (PAPSA) was overturned, allowing states to legalize sports betting. The first state outside of Nevada to legalize sports betting was Delaware on June 5, 2018. As of today, sports betting has been legalized in 39 states, Washington, D.C., and Puerto Rico. Much of this expansion has occurred through the emergence of the online sports betting industry, which has experienced rapid growth as sportsbooks such as FanDuel, DraftKings, and Caesars pursued aggressive market expansion strategies. Constant access to gambling via online apps has clearly captured the attention of millions of people.

The pervasiveness of sports betting as a new pastime among Americans has been substantial. According to the American Gaming Association, sports betting gross gaming revenue through 2025 has totaled \$16.96 billion, representing an 22.8% increase since the previous year.¹ Since 2018, the total amount wagered has approached \$450 billion², with no clear signs of slowing, as legalization has yet to occur in 11 states. Survey evidence from Siena University suggests that approximately 22% of Americans hold an account with at least one online sportsbook.³

This expansion has not been without consequence. For most individuals, online sports betting represents a recreational activity done socially while watching sports. However, the increased ease of access to online gambling has created a largely frictionless environment for ”problem gamblers” to engage in financially destructive behavior. The perception that sports betting is something that can be ”won” is often misguided, as outcomes are largely driven by chance rather than any particular patterns. Broad legalization has therefore facilitated greater exposure to financial danger as more consumers take on greater risk when gambling.

¹<https://www.americangaming.org/resources/commercial-gaming-revenue-tracker/>

²<https://rg.org/statistics/us>

³<https://sri.siena.edu/2025/02/18/22-of-all-americans-half-of-men-18-49-have-active-online-sports-betting-account/>

This paper studies how the legalization of online sports betting has affected financial health using data from the New York Federal Reserve’s Consumer Credit Panel (CCP). The CCP provides annual, state-level measures of debt, and delinquency for all U.S. states, Washington, D.C., and Puerto Rico. Exploiting the staggered rollout of sports betting legalization across states, we apply the difference-in-differences framework developed by Callaway & Sant’Anna (2021) to estimate the Average Treatment Effect on the Treated (ATT). Our results indicate that legalization has led to modest but statistically meaningful deteriorations in financial health. Specifically, we find increases of approximately 0.4 percentage points in credit card delinquency rates, 0.17 percentage points in auto loan delinquency rates, and roughly 1.1 percentage points in the unemployment rate. Overall, the findings suggest that increased access to online sports betting has adversely affected financial stability.

2 Literature Review

A recent study by Hollenbeck et al. (2024) examines the impact of sports betting legalization on financial health, and serves as a primary motivation for this paper. Using data from the University of California Consumer Credit Panel (UC CCP), the authors analyze changes in credit scores and other financial indicators, employing the methodology developed by Callaway & Sant’Anna (2021). They find that credit scores decline by approximately 0.3% in states that legalize sports betting, driven largely by increased debt burdens, and reduced access to credit. Although the UC CCP allows more detailed analysis, we can approximate Hollenbeck’s approach.

This paper also relates to Baker et al. (2024), who studies the effects of legalized sports betting using household-level data. They find that households engaging in sports betting experience an average annual income loss of \$1,100, and increased debt balances, with the strongest effects concentrated among lower-income households. Their analysis employs a two-way fixed effects (TWFE) difference-in-differences framework.

However, recent work by Borusyak et al. (2024), and others demonstrates that TWFE estimators may be biased in settings with staggered treatment adoption in part due to negative weighting of comparison groups. To address these concerns, we adopt the estimator proposed by Callaway & Sant’Anna (2021), referred to as the CS estimator in this paper. This approach accounts for treatment timing heterogeneity, and aggregates cohort-specific effects into an overall ATT.

While our focus is on financial outcomes, related research highlights important health implications of gambling expansion. Taylor et al. (2024) document increases in problematic gambling behavior following online sports betting legalization, including a 150% rise in calls to the National Council on Problem Gambling hotline over a five-year period. These findings underscore that the consequences of sports betting legalization extend beyond financial outcomes to broader measures of well-being.

3 Data

Our primary financial data is drawn from the New York Federal Reserve’s Consumer Credit Panel (CCP), which contains credit records from a 5% nationally representative sample of Equifax credit filings. The data is available annually at the state level, including Washington, D.C., and Puerto Rico, and span the years 2003–2024. Key variables include the share of credit card, auto loan, mortgage, and student loan balances that are 90 days or more past due, as well as total debt balances by category.

Demographic data are obtained from the Bureau of Labor Statistics (BLS), including annual state-level population, unemployment rates, and labor force measures. Economic variables are sourced from the Bureau of Economic Analysis (BEA), including real gross domestic product (GDP), and real income per capita. Both datasets cover the period from 2005 through 2024.

For now, the policy ”implementation year” is defined as 2018.⁴ To focus on relevant pre-

⁴Not technically implemented this year. Referring to the Supreme Court ruling in *Murphy v. NCAA*.

treatment dynamics, we restrict the sample to observations from 2012 onward. Puerto Rico is excluded due to incomplete CCP coverage in recent years⁵. Because this study focuses on the effects of online sports betting, we exclude states that permit only retail (offline) betting, including Washington, New Mexico, North Dakota, South Dakota, Nebraska, Wisconsin, and Mississippi. Including these states as untreated could bias estimates as we are specifically looking to see if online sports betting affects financial outcomes. Nevada is also excluded, as sports betting was legalized prior to the study period.

Several states adopted online sports betting only near the end of the sample period (e.g., Delaware, Maine, Florida, Vermont and Kentucky)⁶, limiting the availability of post-treatment observations. These states are classified as untreated. In total, eight states are removed from the sample, and four states are classified as untreated despite legal adoption due to insufficient data. The final sample consists of 43 states, and Washington, D.C., of which 26 experience treatment during the sample period. Table 1 lists the treated states, and their respective legalization years.

Below, Table 2 & Table 3 contain summary statistics states prior to receiving treatment, and states that have received treatment respectively. Though we have not yet run our model, we can already note that states that do legalize sports betting experiences larger indicators of debt. While demographic, and economic indicators do show growth, this is a result of the fact that these indicators grow over the course of time. Therefore, they would be greater in the post-treatment states where the time-frame is further along.

4 Methodology

The main variable of interest is the credit card delinquency rate. The reason we are using this as our proxy for financial health, other than lack of access to other data, is that it is an indicator of stress on short-term financing. When a consumer uses a credit card,

⁵Missing data from 2017-Present.

⁶Legalized post-September 2023. Would functionally only have a part of a single period of post-treatment.

Table 1: Treatment Start Dates by State

	State	First Start	Online	Offline
1	New Jersey	Jun 2018	Aug 2018	Jun 2018
2	West Virginia	Aug 2018	Aug 2018	Aug 2018
3	Pennsylvania	Nov 2018	May 2019	Nov 2018
4	Rhode Island	Nov 2018	Sep 2019	Nov 2018
5	Arkansas	Jul 2019	Mar 2022	Jul 2019
6	New York	Jul 2019	Jan 2022	Jul 2019
7	Iowa	Aug 2019	Aug 2019	Aug 2019
8	Indiana	Sep 2019	Oct 2019	Sep 2019
9	Oregon	Oct 2019	Oct 2019	
10	New Hampshire	Dec 2019	Dec 2019	Aug 2020
11	Illinois	Mar 2020	Jun 2020	Mar 2020
12	Michigan	Mar 2020	Jan 2021	Mar 2020
13	Montana	Mar 2020	Mar 2020	Mar 2020
14	Colorado	May 2020	May 2020	May 2020
15	District of Columbia	May 2020	May 2020	Jul 2020
16	Tennessee	Nov 2020	Nov 2020	
17	Virginia	Jan 2021	Jan 2021	
18	North Carolina	Mar 2021	Mar 2024	Mar 2021
19	Arizona	Sep 2021	Sep 2021	Sep 2021
20	Wyoming	Sep 2021	Sep 2021	
21	Connecticut	Oct 2021	Oct 2021	Oct 2021
22	Louisiana	Nov 2021	Jan 2022	Nov 2021
23	Maryland	Dec 2021	Nov 2022	Dec 2021
24	Kansas	Sep 2022	Sep 2022	Sep 2022
25	Massachusetts	Jan 2023	Mar 2023	Jan 2023
26	Ohio	Jan 2023	Jan 2023	Jan 2023

Table 2: Summary Statistics for Pre-Treatment States

	Min	Median	Mean	Max	SD	N
Debt Indicators						
Credit Card Delinquencies	3.98	7.45	7.794	16.86	1.732	430
Credit Card Balance	1910	2930	3021.81	5170	607.75	430
Auto Loan Delinquencies	1.64	3.405	3.662	9.79	1.361	430
Auto Loan Balance	2280	4245	4366.63	7920	1009.84	430
Mortgage Delinquencies	0.16	1.29	1.808	15.94	1.704	430
Mortgage Balance	14620	31075	33808.37	68120	11734.03	430
Student Loan Delinquencies	0.13	10.21	9.497	18.36	4.185	430
Student Loan Balance	2690	4910	5045.72	13420	1320.20	430
Economic & Demographic Indicators						
Real GDP	31494.6	239707.4	435657.3	3306929	570210.7	430
Labor Force	292718	2187556	3554621	1.96e+07	4045472	430
Unemployment Rate	1.90	4.80	5.083	11.60	1.779	430
Income per Capita	34812	49376	51384.86	86232	10530.81	430
Population	442898	3783189	5675942	3.16e+07	6507929	430

Table 3: Summary Statistics for Post-Treatment States

	Min	Median	Mean	Max	SD	N
Debt Indicators						
Credit Card Delinquencies	5.74	8.17	8.469	12.90	1.577	129
Credit Card Balance	2210	3410	3467.29	5360	711.70	129
Auto Loan Delinquencies	1.89	3.84	4.116	12.82	1.765	129
Auto Loan Balance	3280	4940	5026.82	6890	767.02	129
Mortgage Delinquencies	0.22	0.59	0.650	2.00	0.337	129
Mortgage Balance	15430	36390	38834.96	80970	14036.20	129
Student Loan Delinquencies	0.12	0.95	3.352	18.01	3.789	129
Student Loan Balance	3530	5880	5935.97	13600	1583.60	129
Economic & Demographic Indicators						
Real GDP	37182.6	368057.0	415581.5	1840059	384218.6	129
Labor Force	289152	3146263	3130557	9862815	2413304	129
Unemployment Rate	2.20	3.90	4.426	10.00	1.674	129
Income per Capita	41001	64464	66777.35	111185	13074.49	129
Population	453649	4627420	5016545	1.63e+07	3934269	129

more often than not, it is to finance an immediate purchase. Auto loans, student loans, or mortgage loans, which are typically paid off in multiple installments over the long term. In comparison credit cards are a shorter-term form of debt that is typically paid off in one payment. Therefore, if financial institutions are reporting higher levels of credit card delinquencies, this indicates that consumers are struggling even to pay off shorter-dated debt indicator that their financial health may become more strained.

We are utilizing real income per capita, population levels, unemployment rates, and real gross domestic product (GDP) as our covariates. Real income per capita functions as a means to capture the differences in average incomes across states. Income levels can reflect a consumer's ability to pay off debt; higher incomes could reduce likelihoods of delinquency, and mean greater credit access. Higher population levels can mean that a given state has more people using credit, which can affect delinquency rates. The unemployment rate measures how many people in the labor force do not have access to a source of income in the form of wages. Higher unemployment rates would reflect higher delinquency rates, as consumers would not only lack the ability to pay off debt but would also likely have to take on additional debt to finance day-to-day transactions. Real GDP is a measure of consumption spending, and of economic growth in a given state. This can therefore isolate the effect of sports betting legalization from economic conditions. We are taking the log of real GDP to report on percentage changes, which measure economic growth.

Additionally, we are also controlling for auto loan delinquencies, mortgage delinquencies, auto loan debt balances, and mortgage balances. As longer-term indicators of financial stress, these covariates should capture the effects of legalized sports betting on long-term financial health, which credit card debt alone does not. We are explicitly not including student loan debt indicators as government policy temporarily paused (and subsequently unpaused) student loan payments which could introduce bias into the model. Credit card delinquency rates can also be affected by people signing up for new credit card and allowing their balances to become delinquent. Because of this, we will also control for the changes

(ln) in credit card debt balance when estimating our ATT for credit card delinquencies. The CCP tracks the same 5% sample of Equifax filings, so changes in debt balances should be adequate in capturing the effect of new sign ups. However, our baseline model will not include this control since it is more of an approximation of this effect given limitations in our data.

We will implement a difference-in-differences (DID) identification to determine our relationship. As mentioned before, we will be utilizing the procedure proposed by Callaway & Sant’Anna (2021), which we refer to as the CS estimator.

$$ATT_{dr}^{ny}(g, t) = \mathbb{E} \left[\left(\frac{G_g}{\mathbb{E}[G_g]} - \frac{\frac{p_{g,t}(X)(1-D_t)}{1-p_{g,t}(X)}}{\mathbb{E} \left[\frac{p_{g,t}(X)(1-D_t)}{1-p_{g,t}(X)} \right]} \right) (Y_t - Y_{g-1} - m_{g,t}^{ny}(X)) \right] \quad (1)$$

$$where \ m_{g,t}^{ny}(X) = \mathbb{E} [Y_t - Y_{g-1} \mid X, D_t = 0, G_g = 0] \quad (2)$$

This is the AIPW/Doubly-Robust estimand, in which we exploit heterogeneous treatment timings to estimate our ATT while using “not-yet-treated” units as our control group. A standard DID model builds a comparison between a treated, and untreated group by taking the difference in average treatment effect before, and after policy implementation. However, this only works to create unbiased estimates if the treatment timing is the same between all groups. Each state legalized sports betting at different times, which means we cannot use standard DID models like TWFE.

Simply put, the CS estimator runs the standard DID procedure of comparing differences between averages, but across every possible period of time between every possible cohort of treated, and not-yet-treated group. For example, it would group states that have been treated with sports betting in 2018, then compares them to our not-yet-treated control group⁷ to create an estimate of the effect of policy implementation. It repeats this for 2017, 2019, 2020, 2021, and so on. It then aggregates all estimates into a single number using a

⁷States that have not experienced legalized sports betting as of 2018.

simple weighted average of all $ATT(g, t)$:

$$\theta_W^{\text{simple}} := \frac{1}{\kappa} \sum_{g=2}^T \sum_{t=2}^T \mathbf{1}\{g \leq t\} ATT(g, t) P(G = g \mid C \neq 1) \quad (3)$$

The aggregated value will be the primary focus of the paper. Additionally, we will be utilizing the "dynamic" (event-time) aggregation method to prove our parallel trends assumption. This equation allows us to capture specific effects of the policy before, and after implementation at certain periods of time:

$$\theta_D^{\text{event}}(e) = \sum_{g=2}^T \mathbf{1}\{g + e \leq T\} ATT(g, g + e) P(G = g \mid G + e \leq T, C \neq 1) \quad (4)$$

From here, we can derive results from our model, and come to conclusions about the effects of the policy implementation. There is a potential issue, however, in that there is less data for states that implemented sports betting later (in 2023), leaving only two periods of post-treatment. This means that our ATT for those later periods may not be unbiased. However, due to the way the CS estimator weighs ATT, these later periods may hold less weight in the final aggregate calculation, mitigating this concern.

5 Results

Before we can move onto running our model, we must prove the parallel trends assumption to show the validity of this study. The parallel trends assumption states that treated groups, and control groups must follow the same path over time if treatment had not occurred at all. Through the "csdid" package provided through Stata, we can do so using built-in commands. Below is a plot of our event study for credit card delinquencies using the `csdid_plot` command. The plot suggests that the parallel trend holds visually.

The ATT prior to policy implementation shows no particular positive, nor negative trend. Most of the calculated ATT pre-treatment hovers around 0, indicating no effects. Addi-

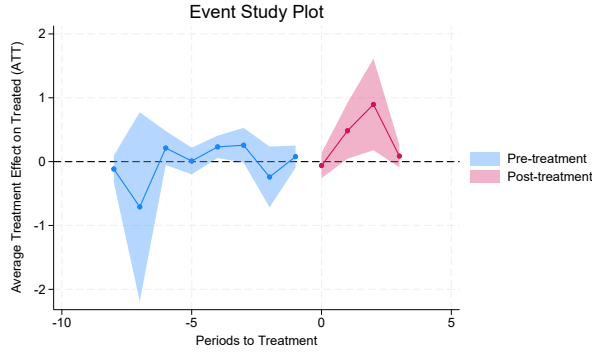


Figure 1: Credit Card Delinquency

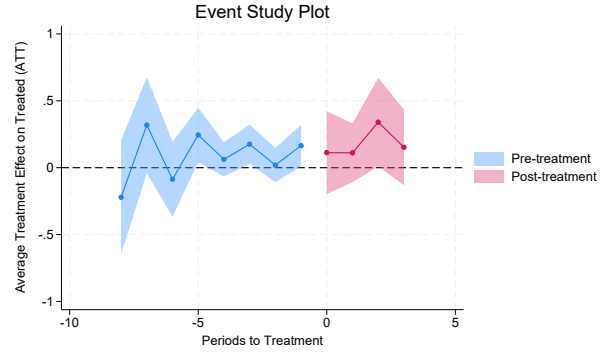


Figure 2: Auto Loan Delinquency

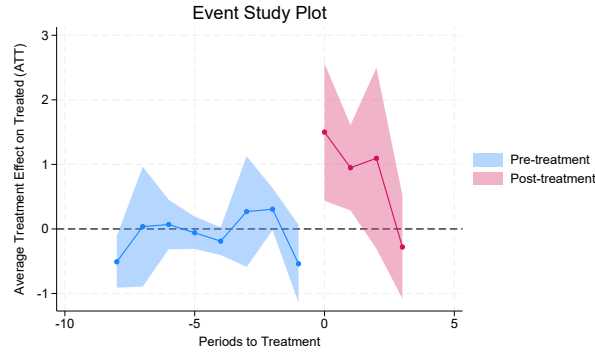


Figure 3: Unemployment Rate

tionally, we can further verify the parallel trends assumption by looking at the raw values generated through Stata in Table 4 below. Our “pre_avg” value is statistically insignificant. This suggests that there are no differing trends between our treated, and control groups prior to the implementation of our policy. Therefore, the assumption holds. This test was also conducted for auto loan delinquencies, and the unemployment rate. Parallel trends still holds for these variables.

Table 4: Event Study Plot: Credit Card Delinquencies

	Coefficient	p-value	95% Confidence Interval
Pre_avg	-0.034314 (0.06976)	0.623	[-0.17104, 0.10241]
Post_avg	0.351012 (0.09492)	0.000	[0.16498, 0.53705]
Tm8	-0.115621 (0.10777)	0.283	[-0.32685, 0.09561]
Tm7	-0.708116 (0.75605)	0.349	[-2.18994, 0.77371]
Tm6	0.212825 (0.13647)	0.119	[-0.05466, 0.48031]
Tm5	0.009392 (0.10651)	0.930	[-0.19937, 0.21815]
Tm4	0.231683 (0.08926)	0.009	[0.05674, 0.40662]
Tm3	0.257301 (0.13861)	0.063	[-0.01436, 0.52896]
Tm2	-0.240385 (0.24327)	0.323	[-0.71718, 0.23641]
Tm1	0.078412 (0.08858)	0.376	[-0.09521, 0.25203]
Tp0	-0.061674 (0.10095)	0.541	[-0.25953, 0.13618]
Tp1	0.482848 (0.22306)	0.030	[0.04566, 0.92004]
Tp2	0.896272 (0.36650)	0.014	[0.17795, 1.61459]
Tp3	0.086602 (0.08980)	0.335	[-0.08940, 0.26261]

Table 5 below contains the results that we derived from our model. As we can see, the ATT on credit card delinquencies is approximately an increase of 0.402 percentage points. Our results are significant at the 1% level. When controlling for changes in credit card debt balances, our ATT is approximately an increase of 0.390 percentage points. This is also significant at the 1% level. Additionally, we determined that auto loan delinquencies rose by approximately 0.2 percentage points, though this result is difficult to accept as interpretable due to the low statistically significant. We also ran our model on the unemployment rate to examine potential deterioration in the labor market. There is a statistically significant 1.1 percentage point increase in the unemployment rate in states that legalize sports betting.

Overall, these results align with the findings of Hollenbeck et al., 2024, and Baker et al., 2024, who also discovered deterioration in financial health following the legalization of sports betting. As of now, sports betting is being rapidly deregulated across the nation, seemingly with little regard for the financial implications. Although sports betting generates significant economic value for a state in the form of additional taxes (bringing in approximately \$2,838,258,264 in state tax revenue in 2024)⁸, it has been a detriment to households. This is just in the first few years of broad sports betting legalization. While there exist regulations to limit customer access to sportsbooks, it is clearly not enough. Broader regulation is necessary to prevent this gambling epidemic from worsening.

Prior to estimation, we assess multicollinearity among covariates using variance inflation factors (VIF), and find no evidence of problematic multicollinearity, with all VIF values below 5, indicating stable coefficient estimates. These values can be seen in Table 6 below. Standard errors are clustered at the state level to account for within-state serial correlation in the panel data. Inference is conducted using a weighted bootstrap with 999 replications, which repeatedly re-estimates the model to obtain robust standard errors, and confidence intervals.

⁸<https://rg.org/statistics/us>

Table 5: Average Treatment Effect on the Treated

	No Controls	Baseline	Debt Controls
Credit Card Delinq.	0.023 (0.087)	0.402*** (0.108)	0.390*** (0.105)
Auto Loan Delinq.	-0.062 (0.079)	0.169 (0.120)	-
Unemployment Rate	0.343* (0.194)	1.061*** (0.224)	-
Controls	No	Yes	Yes
Control Debt	No	No	Yes

Notes: Standard errors are in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 6: Variance Inflation Factors (VIF)

Variable	VIF	1/VIF
Population	3.99	0.251
Log Real GDP	3.93	0.254
Income per Capita	2.52	0.397
Mortgage Balance	2.17	0.461
Auto Loan Balance	2.04	0.489
Mortgage Delinquencies	1.75	0.573
Unemployment Rate	1.69	0.592
Auto Loan Delinquencies	1.22	0.819
Mean VIF	2.41	

6 Conclusion

Sports betting is actively a dangerous activity for many. There is no denying that many people who engage in sports betting do so responsibly. However, stricter regulations are not meant to address this cohort. For people who already have a gambling issue, or are susceptible to gambling addiction, the surge in online sports betting is an epidemic. No matter where you look, dozens of ads pushing for people to sign up for FanDuel, BetMGM, or any of the other sportsbooks are there. And these companies are finding the exact customer base they are looking for in addicts. There used to be some level of friction in gambling.

If you wished to indulge yourself, you would have to travel to Las Vegas, or Atlantic City. As things currently are, access to gambling is essentially frictionless. It is up to policy to address this ever-growing epidemic. If no action is taken by regulators (perhaps in the form of stricter state side regulation), this increase in sports betting will almost certainly lead to greater financial strain for many consumers as it already has.

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